



SIMATS Online Programming Lab

DS PROGRAMMING



YUGITHA B
192424102RD



QUESTIONS

10 / 10 completed

Binary Search - Array

BINARY SEARCH - ARRAY

Write a C program to perform Binary Search on a sorted array of 10 integers. The program should accept the array elements and a target value from the user. It must display the position of the element and the total number of comparisons made.

Sample Input:

Enter 10 sorted integers:

5 12 18 23 34 45 56 67 78 90

Enter element to search: 45

Sample Output:

Element 45 found at position 6

Comparisons made: 3

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[10];
5     int target;
6     int left = 0, right = 9;
7     int comparisons = 0;
8     int foundIndex = -1;
9
10    printf("Enter 10 sorted integers:\n");
11    for (int i = 0; i < 10; i++) {
12        scanf("%d", &arr[i]);
13    }
14
15    printf("Enter element to search: ");
16    scanf("%d", &target);
17
18    while (left <= right) {
19        int mid = left + (right - left) / 2;
20        comparisons++;
21    }
```

OUTPUT

```
Enter 10 sorted integers:
Enter element to search: Element 45 found at position 6
Comparisons made: 3
```

INPUT

5 12 18 23 34 45 56 67 78 90
45

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